

PROPOSED METHOD

MC-ESO

Multi-Channel Epidemic Spread Optimizer

A population-based black-box optimizer inspired by
multi-route epidemic transmission

Kosei Matsuzaki

Outline

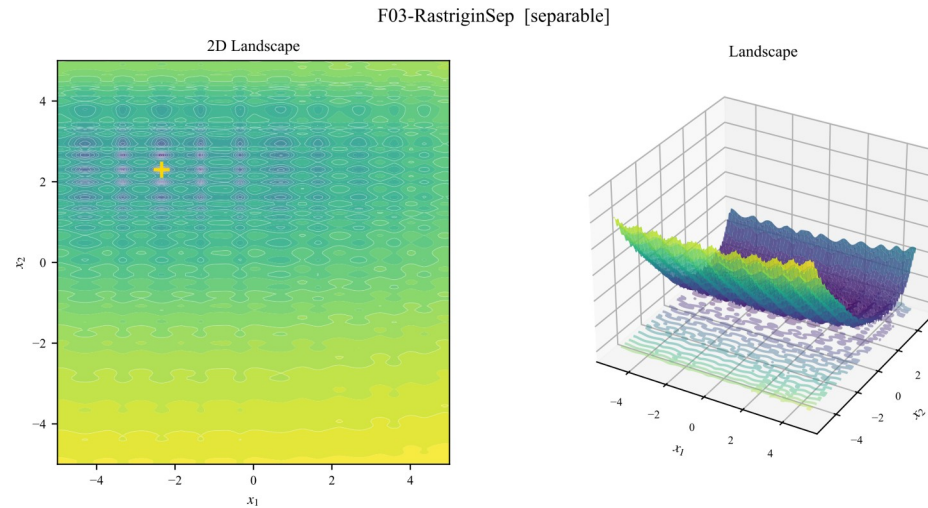
- 1 Background
- 2 Purpose & Significance
- 3 Proposed Method
- 4 Experiments
- 5 Results
- 6 Discussion
- 7 Conclusion

1

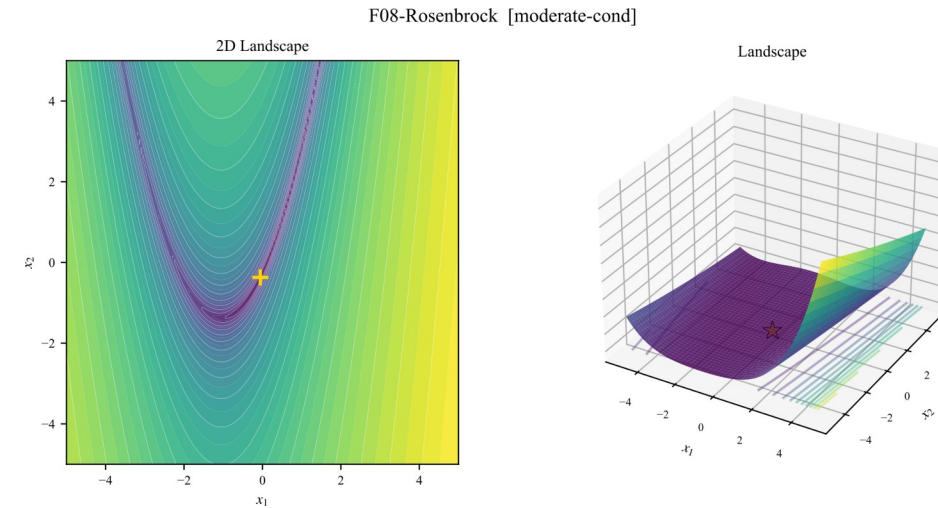
Background

The shared weakness of existing methods, and what motivated this work

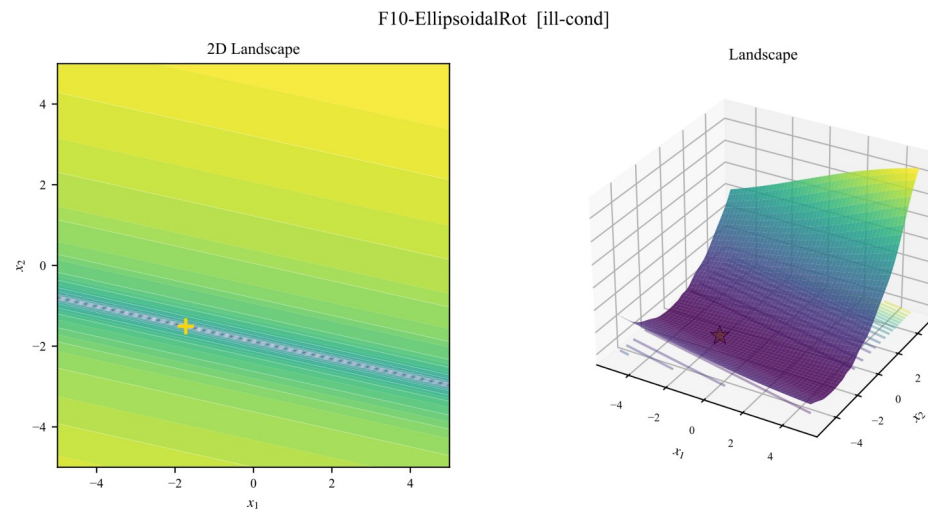
Black-box functions come in many flavours of hard



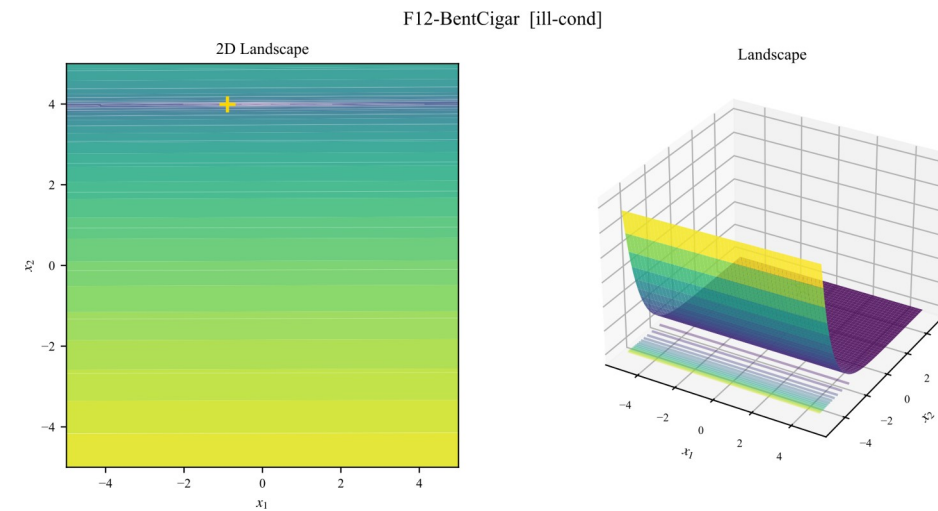
Multimodal — many local optima



Narrow curved valley



Ill-conditioned & rotated

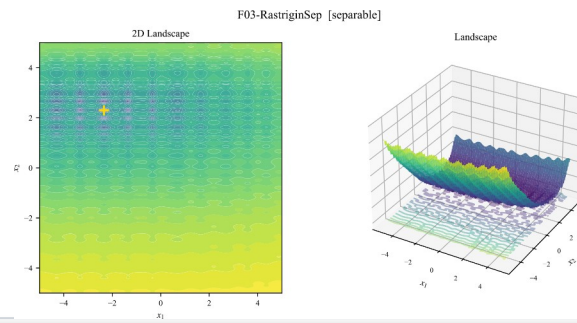


Extreme anisotropy

Every existing method has a class of problems it can't solve

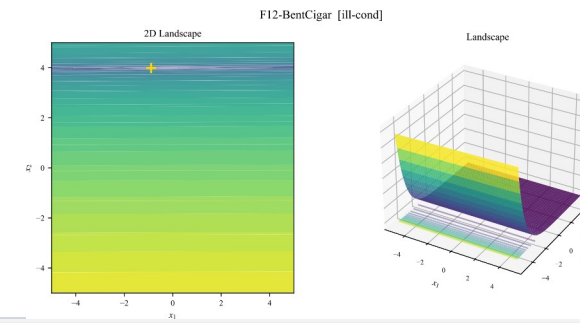
CMA-ES

Gets trapped in local optima on multimodal landscapes



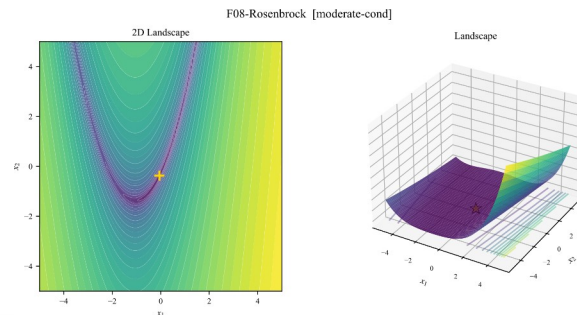
DE

Struggles when the problem is strongly anisotropic



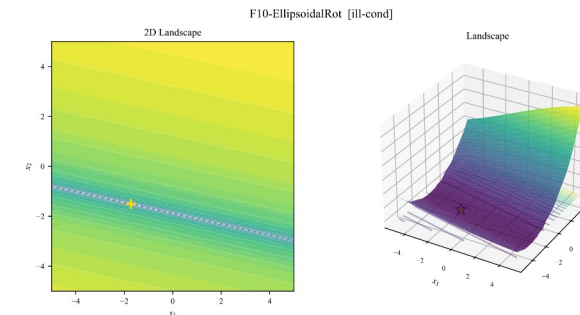
PSO

Stagnates on narrow valleys

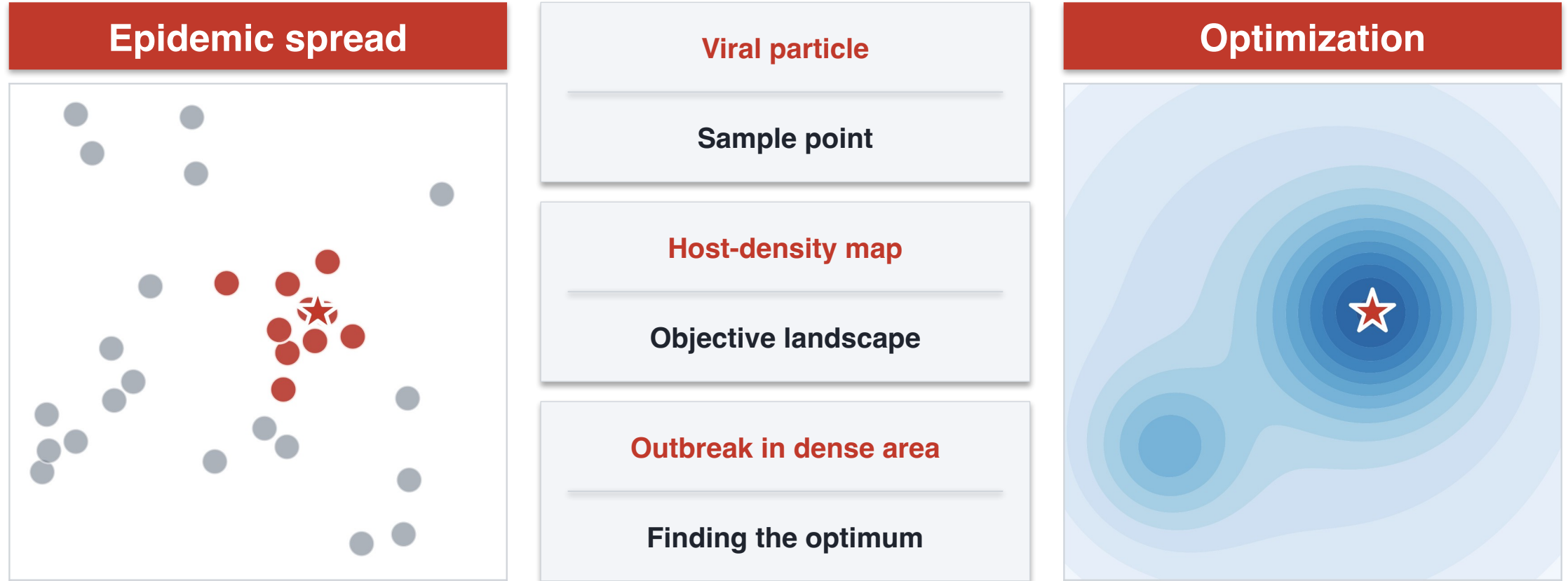


SaVOA

Single channel — can't both escape AND refine



Epidemic spread and optimization share the same structure



→ Borrow how an epidemic concentrates on dense regions

2

Purpose & Significance

What we ask, and why it matters

Goal & contributions

GOAL

Build a search method that performs reasonably on every function

Novelty

First metaheuristic to combine multiple transmission routes (epidemic-inspired)

Mechanism

Channel mixing + population control + step-size adaptation

Outcome

High success rate across the BBOB benchmark — see next slide

Result preview — overall and per-category SR@1e-4

26 BBOB functions × 10 runs each — higher = solves more functions

MC-ESO 91.9%	DE 91.9%	PSO 73.1%	SaVOA 73.1%	CMA-ES 71.2%
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Category	MC-ESO	DE	PSO	SaVOA	CMA-ES
Separable	100%	98%	98%	82%	72%
Moderate cond.	98%	95%	100%	100%	93%
Ill-conditioned	94%	100%	40%	52%	100%
Multimodal	92%	98%	80%	68%	44%
Weak structure	74%	66%	42%	58%	40%

→ MC-ESO is the only method that stays $\geq 74\%$ in every category

3

Proposed Method

MC-ESO — generation flow, three channels, three population mechanisms

One generation: 5 steps, each with its own innovation

1 Spawn

Three transmission channels

Contact / Droplet / Airborne in parallel

2 Evaluate

(no new mechanism)

Fitness call only

3 Update strain pool

Strain coexistence

Keep up to 6 spatially-separated elites

4 Select population

Host competition

$\mu + \lambda$ greedy with parent rollback

5 Restart?

Spillover + σ adaptation

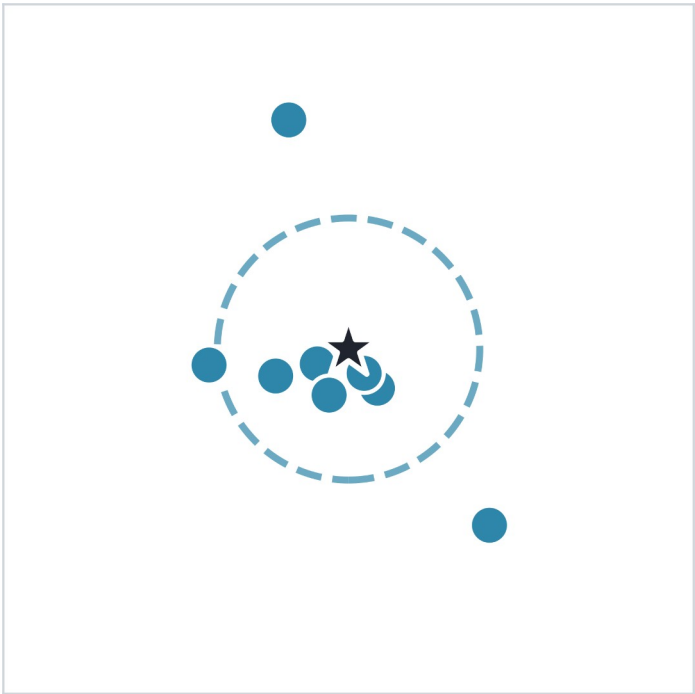
Escalating re-seed when stagnated

→ The next slides detail each innovation (rows 1, 3, 4, 5)

Three transmission channels

Close-contact

p = 30 %

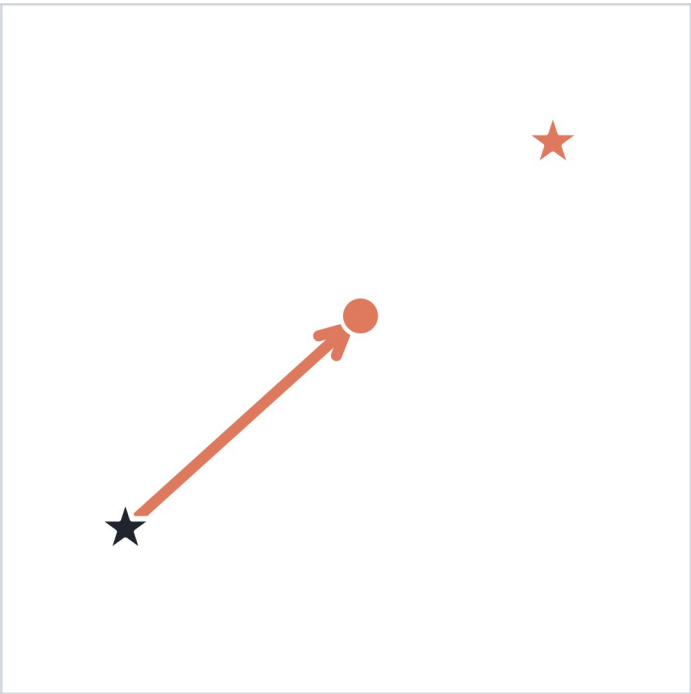


Mimics: Person-to-person touch

Spawn near the source — exploit the local basin

Droplet

p = 40 %



Mimics: Coughs / sneezes carry the virus to neighbours

Pull toward another elite strain — implicit anisotropy

Airborne

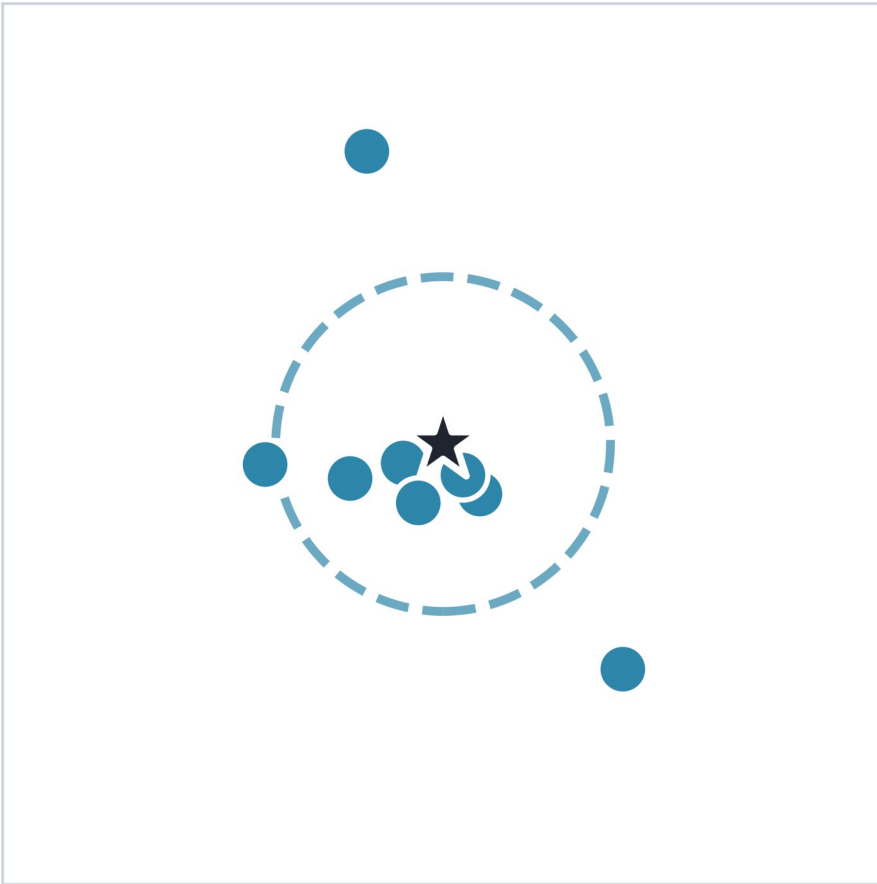
p = 30 %



Mimics: Aerosols drift far from any source

Spawn far from the population — escape when stuck

Channel — Close-contact



Close-contact

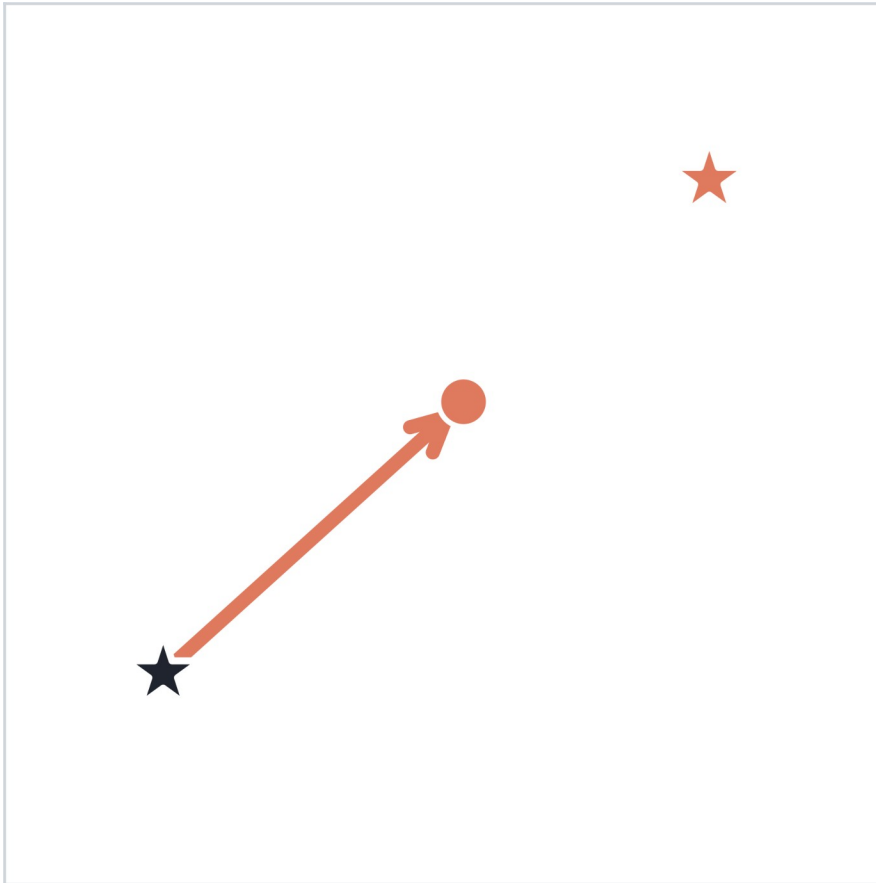
FORMULA

$$\mathbf{x}_p + \mathcal{N}(0, \sigma_i \mathbf{I})$$

ROLE

1. Pick a parent $\mathbf{x}_p$ from the population
2. Sample a Gaussian step with width σ_i
3. σ_i shrinks with fitness and parent age
4. → Refines the basin the parent sits in

Channel — Droplet



Droplet

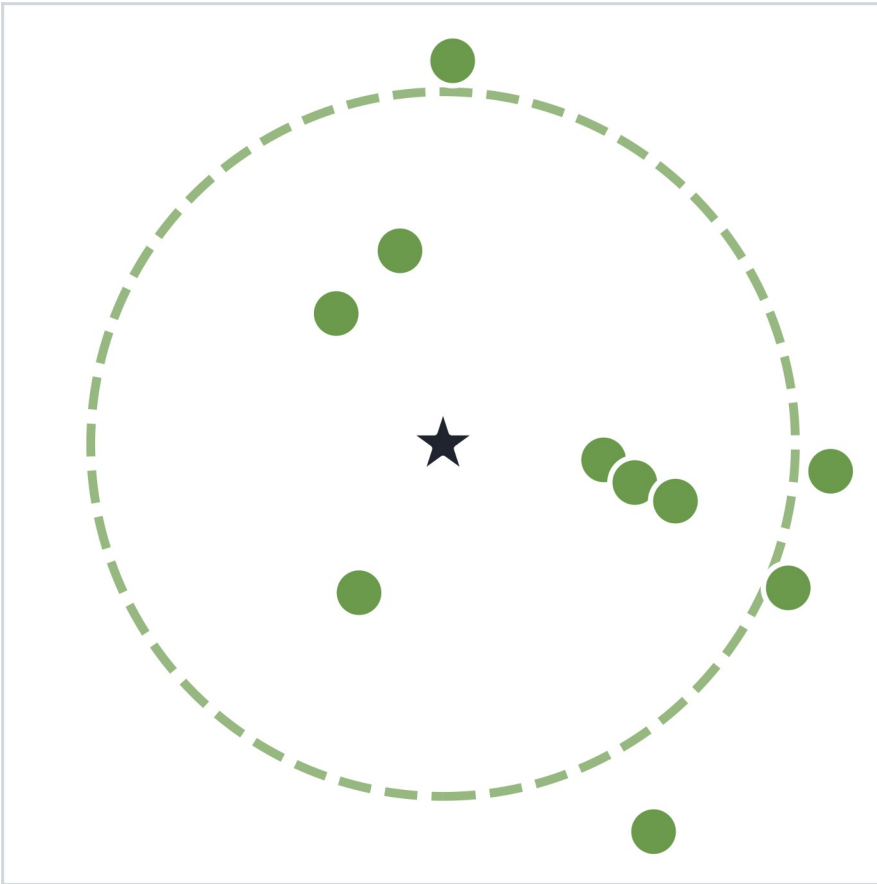
FORMULA

$$\mathbf{x}_p + F(\mathbf{x}_{\text{strain}} - \mathbf{x}_p) + F(\mathbf{x}_a - \mathbf{x}_b)$$

ROLE

1. Pull the parent toward another elite strain
2. Perturb with a population differential vector
3. No covariance matrix is learned explicitly
4. → Implicit anisotropy — alignment is automatic

Channel — Airborne



Airborne

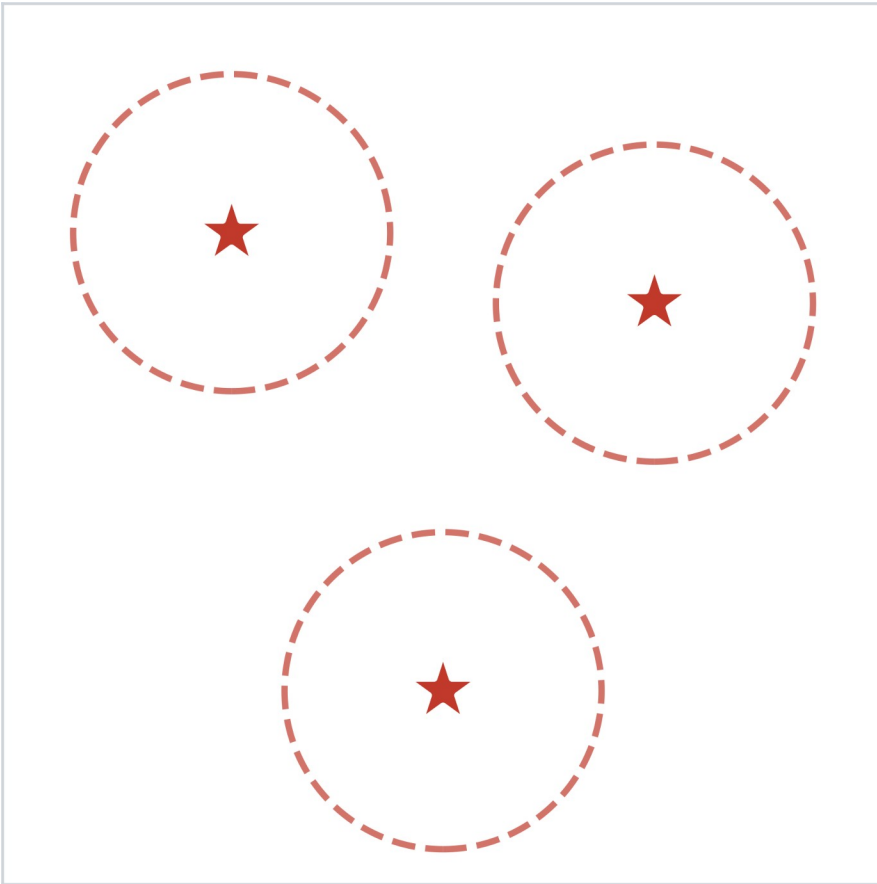
FORMULA

$$\mathbf{x}_{\text{rand}} + \mathcal{N}(0, \sigma_{\text{air}} \mathbf{I})$$

ROLE

1. Sample a random location, independent of any parent
2. Use a wide step width σ_{air} (much larger than σ_i)
3. Independent of where the population currently sits
4. → Escape route when the swarm is trapped

Population mechanism — Strain coexistence



Strain coexistence

Keep up to 6 spatially-separated elite strains

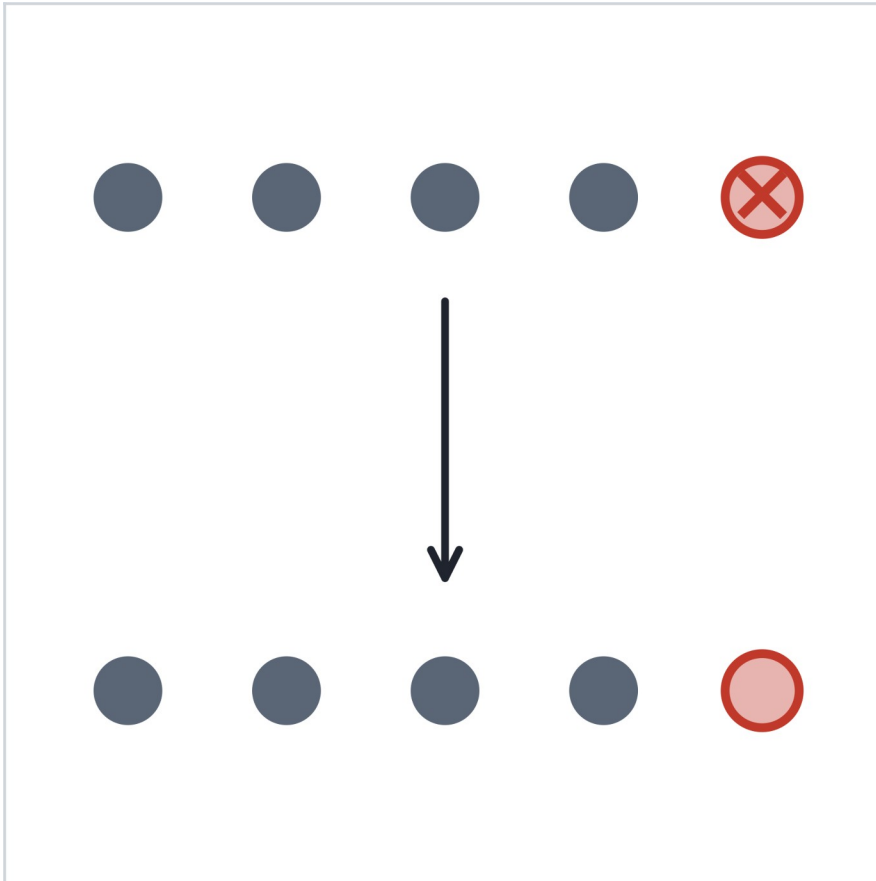
Instead of locking onto a single source, we protect several elite strains that are spaced apart.

→ The droplet channel can pull toward any of them, so the search

stays diverse on multimodal landscapes.

```
niche_radius = 1.0    n_elite_max = 6
```

Population mechanism — Host competition



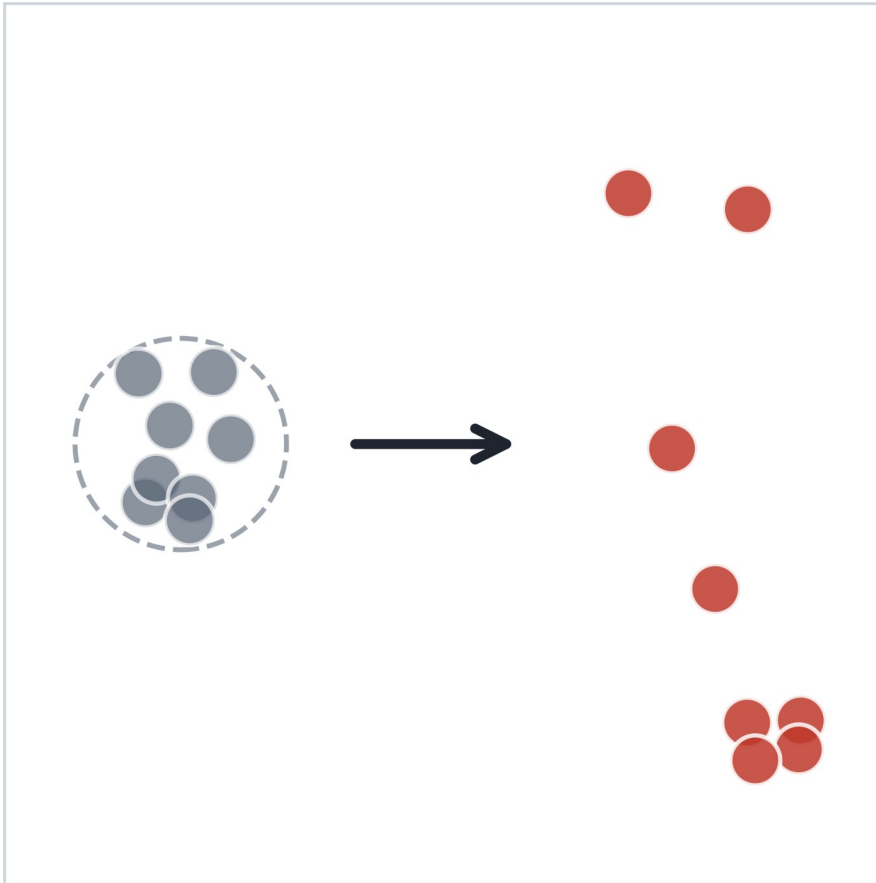
Host competition

Each generation: kill the bottom 25%, then rollback if children are worse

Kill the worst quartile of the population. Insert new children into the empty slots — but if a child is worse than its parent, revert.
→ The population improves monotonically; no regression allowed.

```
kill_fraction = 25%
```

Population mechanism — Spillover



Spillover

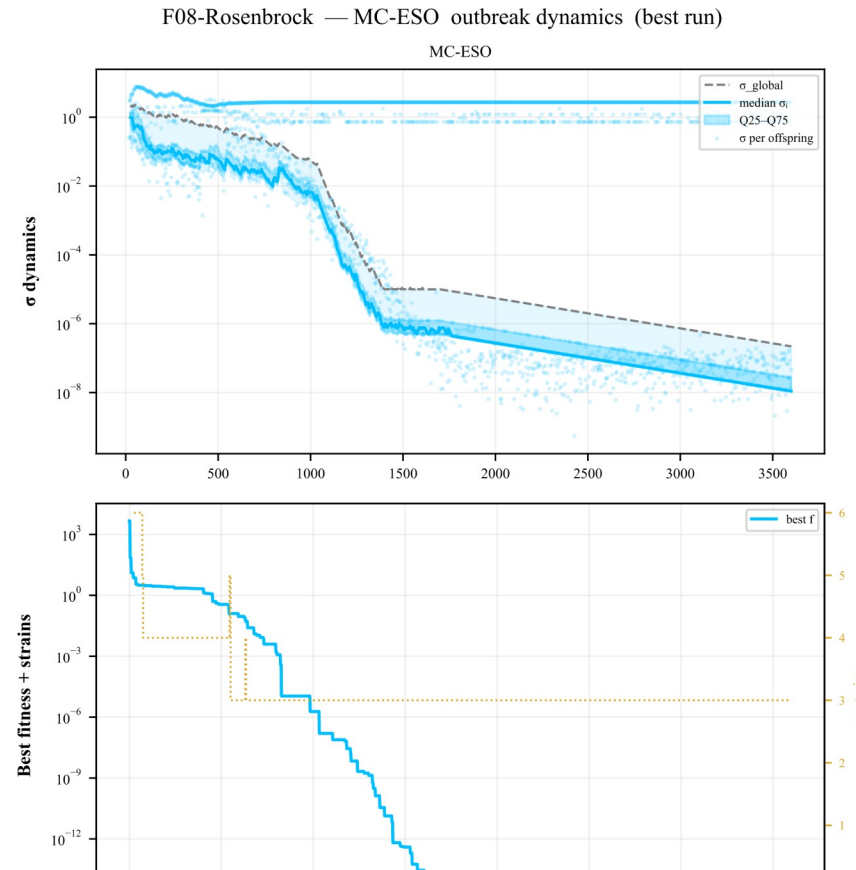
Escalating re-seed when improvement stalls

Reset strength grows with the stagnation streak:

- Stage 0: replace 75% of the population uniformly
- Stage 1: replace 100% (best preserved)
- Stage 2: also discard the best, reset σ (basin switch)

```
no_improve ≥ 300  AND  f_best > 1e-8
```

Step-size σ adaptation



Example: σ trajectory of one MC-ESO run on F08-Rosenbrock

Expand on improvement, shrink on stagnation, drill hard at the end.

4

Experiments

BBOB benchmark — conditions and baselines

Experimental setup

Benchmark	BBOB (24 functions) + 2 classical (C01–C02)
Dimension	$d = 2$, search domain $[-5, 5]^d$
Budget	5,000 evaluations per run
Repetitions	50 independent runs per (function, method)
Success	Run succeeds if best $f \leq$ target threshold ($1e-1 \dots 1e-10$)
Metric	$SR@1e-k$ = success rate at threshold 10^{-k}

Comparison methods (details in appendix)

MC-ESO

Three transmission channels (contact + droplet + airborne) in parallel

CMA-ES

Learns a covariance matrix; reshapes sampling to fit the landscape

PSO

Particles move toward their own best and the swarm's best

SaVOA

Virus-inspired (single channel) with self-adaptive step size

5

Results

Success rate across all functions; the hard cases

Average success rate over 26 BBOb functions

26 functions × 50 runs × SR@1e-4 (= fraction reaching target precision)



→ MC-ESO leads every baseline by at least 19 percentage points

Full BBOB — success rate at multiple precision thresholds

SR @ 1e-2	SR @ 1e-4	SR @ 1e-7	SR @ 1e-10
2472 / 2600	2432 / 2600	2290 / 2600	2184 / 2600
95.1%	93.5%	88.1%	84.0%

Top method on 21 / 26 functions (6 outright + 15 tied)

Where MC-ESO still struggles

The 3 weakest cases — all of them are open problems for every method

Function	SR @ 1e-4	Type	Why it's hard
F24 Lunacek bi-Rastrigin	16%	Double funnel	Deeper funnel is the wrong basin (deceptive)
F23 Katsuura	58%	Fractal	Self-similar; no smoothing scale exists
F18 Schaffer F7 (ill)	88%	Ill-cond. multimodal	Many local optima inside a narrow valley
F04 Büche-Rastrigin	92%	Asymmetric multimodal	Dense grid of local optima, asymmetric

These remain open problems across the entire metaheuristic literature

6

Discussion

Why it works, and where it loses

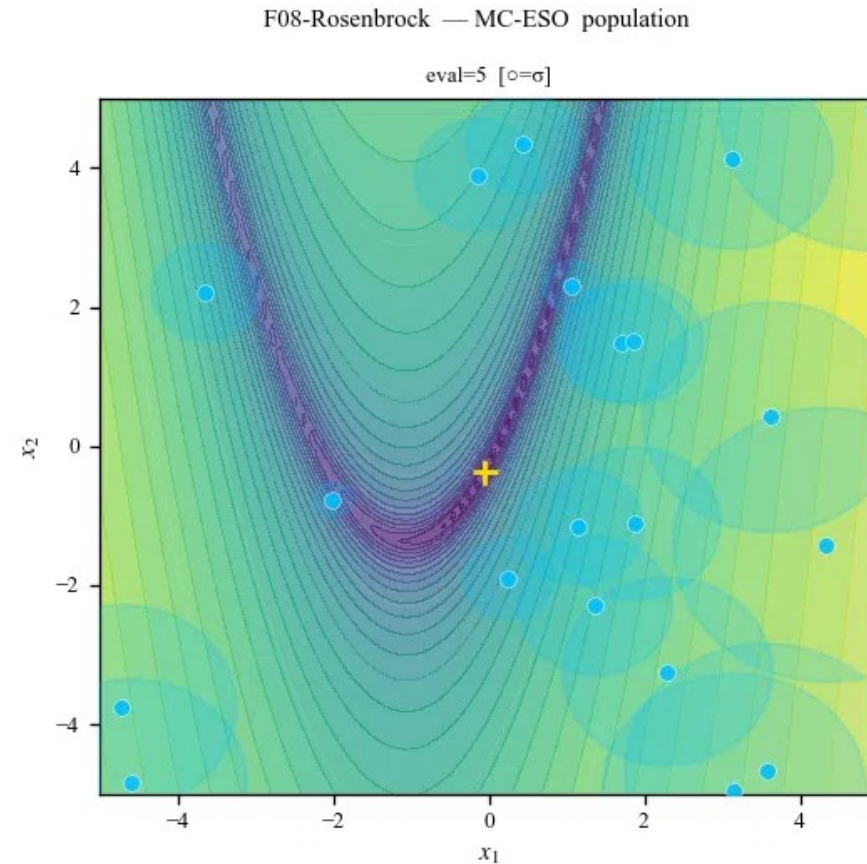
Why does it work — ① Implicit anisotropy

The droplet channel picks up the landscape's anisotropy for free

The differential vector used by the droplet channel naturally aligns with the dominant spread direction of the population.

→ Same benefit as CMA-ES, without explicit covariance.

- Extra cost: zero
- Helps on: narrow valleys like F08 Rosenbrock



From the F08-Rosenbrock run

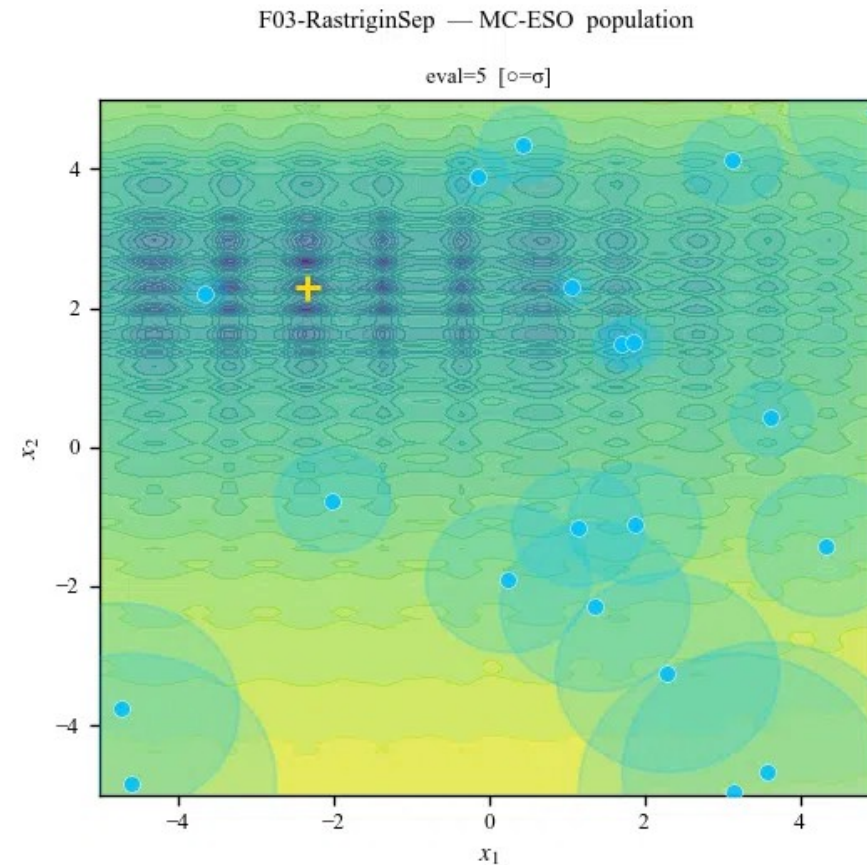
Why does it work — ② Multimodal coverage

Strain coexistence keeps multiple peaks alive simultaneously

Instead of locking onto a single source, MC-ESO keeps up to 6 spatially-separated elite strains.

→ A single trapped optimum no longer terminates the search.

- Extra cost: light (distance check only)
- Helps on: rugged multimodal like F03 Rastrigin



From the F03-RastriginSep run

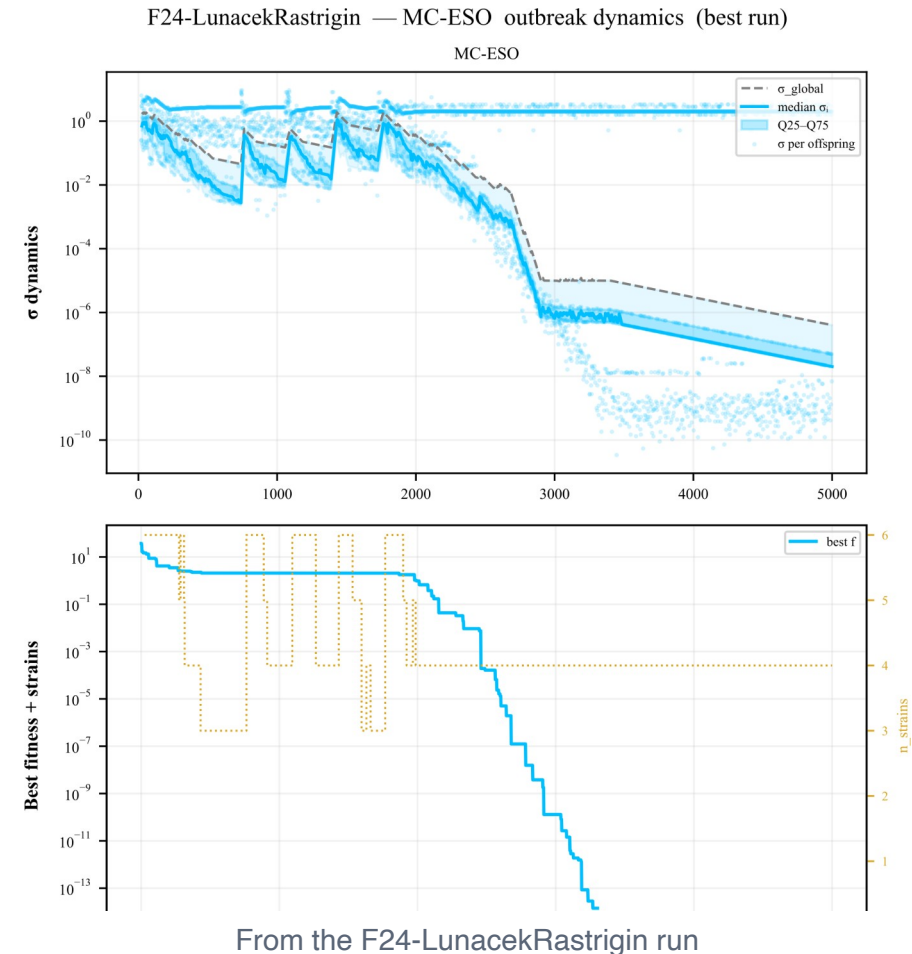
Why does it work — ③ Adaptive disruption

Spillover explosively re-seeds a stuck population

When improvement stalls for long, the reset strength escalates with each successive failure.

→ The algorithm doesn't just wait — it switches basins by itself.

- Helps on: multi-basin landscapes like F24



Limitations

- Hand-tuned channel ratios (30:40:30)
- Deceptive funnels (F24, F23)
- Only $d = 2$ reported
- No gradient hybridization

7

Conclusion

Summary

Conclusion

Idea

Multi-route reproduction inspired by epidemic spread

Mechanism

3 transmission channels + 3 population mechanisms + adaptive σ

Result

Average SR = 93.5% on BBOB-26 — outperforms all 3 baselines

Thank you.

Appendix — SaVOA (Self-adaptive Virus Optimization)

The closest prior work — also virus-inspired, but with a single channel

Inspiration	Viral infection and self-replication
Reproduction	Single Gaussian sample around the parent (contact channel only)
σ update	Self-adaptive — each agent learns its own step size
Population	Simple generational replacement. No niching, no escalating restart.
vs. MC-ESO	• 1 channel $\rightarrow$ 3 (contact + droplet + airborne) • Adds 3 population mechanisms (strain coexistence, host competition, spillover)

Appendix — Other baselines

CMA-ES

Covariance Matrix Adaptation Evolution Strategy (de facto standard)

- Learns the shape of the sampling Gaussian itself
- Excellent on elongated / rotated landscapes
- Tends to get trapped in local optima on multimodal functions

PSO

Particle Swarm Optimization

- Each particle moves toward its own best and the swarm best
- Strong on smooth problems / stagnates on high-dim narrow valleys